Oracle
Retrained on Homology-Restricted MegaScale
Zero-shot
Supervised, not trained on Megascale

Rectified gain @ k=96 (kcal/mol)
100
10
1
0
27.80
14.81
3.38
17.69
9.14
1.62
19.10
9.77
1.88
14.26
6.49
1.06
15.78
8.24
1.56
17.04
8.17
1.57
11.27
3.97
0.58
12.56
5.01
0.84
12.39
4.49
0.61
11.84
4.34
0.82
11.39
5.94
1.30

Predicted most stabilizing per domain
10
1
0
50.65
34.30
11.04
51.83
33.61
10.57
51.65
33.54
10.81
54.23
33.45
8.32
48.78
35.99
11.23
49.97
35.17
10.86
61.61
26.70
7.70
54.30
32.65
9.04
55.22
33.04
7.74
51.65
35.35
9.00
46.43
38.22
11.35

Top 96 per domain
ESM-MSR (p0) WT head
ESM-MSR (p0) combined
MutateEverything st1
ThermoMPNN
SPURS
ESM3-small-open unmasked
ProteinMPNN
ProSST
GeoDDG-3D
Rosetta

% conservation ≤ 25
25 < % conservation ≤ 75
% conservation > 75

Top 96 per Homology-Restricted MegaScale test domain (23 domains with the feature, 27,163 single mutants); gain = sum of max(ddG, 0) over every pick, whatever its predicted sign; boxes: interquartile range with median line; dots: domains; annotations: mean over domains; dashed lines: oracle mean.